INTERNATIONAL TRANSACTIONS ON ELECTRICAL ENERGY SYSTEMS
Int. Trans. Electr. Energ. Syst. 2013; **23**:553–569
Published online 23 February 2012 in Wiley Online Library (wileyonlinelibrary.com). DOI: 10.1002/etep.1612

Market power assessment in electricity markets: supply function equilibrium-based model

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SUMMARY

Because of the oligopoly structure of the electricity markets, the Generation Companies (GenCos) may exert market power by using bidding strategies such as capacity withholding. This brings the market clearing price higher and the quantity generated of the market lower. In this article, capacity withholding in an oligopolistic electricity market in that all GenCos bid in the supply function equilibrium model is analyzed and the capacity withheld index, the capacity distortion index and the price distortion index are formulated. Demand elasticity and forward contracts can significantly affect the GenCos' capacity withholding in the market. The potential ability of market for capacity withholding is measured by the distortion–withheld index proposed in this article. The finding in this article is helpful for market power monitoring and mitigating. A two-settlement market including a forward market and a spot market is used to describe GenCos' strategic forward contracting and spot market competition. Copyright © 2012 John Wiley & Sons, Ltd.

KEY WORDS: capacity withholding; supply function equilibrium (SFE); forward contract; demand elasticity

1. INTRODUCTION

Today it is widely acknowledged that most power markets cannot be modeled under the perfect supposition. In this environment, Generation Companies (GenCos) are able to exercise market power through different bidding strategies. The two components of market power strategy are capacity withholding and financial withholding. Financial withholding is used when GenCos raise their bidding prices at a considerable level to obtain higher profits, and capacity withholding is used when GenCos withhold capacity to push up the market clearing price [1]. However, as regulator approximately knows cost of a specific unit, financial withholding cannot be realized but capacity withholding can cause high-cost units to operate and raise the market clearing price. Hence, it is important to analyze capacity withholding in electricity markets. Newbery have indicated that in the England and Wales power market, GenCos exercise capacity withholding during peak loads to cause more expensive units set market price [2].

It is widely acknowledged that forward contract trading can mitigate market power in the electricity markets. Theoretical analyses of forward and spot market interactions have been presented in many references [3]. Allaz and Vila [4] argued that for a symmetric duopoly, forward trading could make the spot market more competitive. Le Coq and Orzen [5] examined the theoretical result of Allaz and Vila [4] in a controlled laboratory environment and investigated how the trading institution and the number of firms affect competition. In the most of the electricity market models, the forward contracts are taken as given variables [6], [7]. Moreover, there are some contracts in which both the spot and the forward sales of GenCos are taken as decision variables. These contracts are called *strategic forward contracts*. Many of the studies on capacity withholding and market power have not considered the effects of forward contracts or strategic forward contracts on capacity withholding [8–12], [21].

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Demand elasticity is also important for capacity withholding analyzing. Demand elasticity in power markets is low, but a small increase can result in considerable decrease in capacity withholding of GenCos. To date, most of the studies on the strategic bidding behaviors of GenCos considered the demand side as given demand curves [13], [14]. Only a few have considered the effects of demand elasticity on the strategic behavior in the electricity markets [15].

Oligopoly equilibrium models, such as Cournot models, Bertrand models, and supply function equilibrium (SFE) models, play significant roles in analysis of capacity withholding. Among these models, SFE models might be more appropriate for long term analysis [16]. The decision variables in SFE models are the parameters of supply functions rather than quantity and price as in the Cournot and Bertrand models [6]. SFE models are used for power market simulations in [17–20]. In the study of Badlick [14], an example was used to compare Cournot and SFE models of bid-based electricity markets. In the study of Niu *et al*. [6], a linear asymmetric SFE model with transmission constraints is proposed to analyze the bidding strategies with forward contracts.

The indices such as the Herfindahl–Hirschman index (HHI) and the Lerner index have been used in market power analysis. In the study of Gan and Bourcier [22], it has been shown that the HHI cannot reflect local market power. To solve this problem, Yang and Jordan [23] defined and applied system HHI and group HHI. According to regulations set by the Federal Energy Regulatory Council, a market participant can exercise market power only if he or she owns 20% or more of the total market share. In the study of Kanagala *et al*. [24], this benchmark is used as a threshold for market power estimation. Other indices such as the residual supply index [25] and the local market power index [26] are also introduced. Unfortunately, forward contracts are overlooked in these market power indices.

However, three problems in capacity withholding assessment needed to be studied further.

- First, the effects of demand elasticity on capacity withholding needs more study.
- Second, the relation between price distortion and capacity withholding has not been modeled yet.
- Third, the effects of forward contracts on capacity withholding have not been assessed well yet.

In this article, we want to assess the capacity withholding of GenCos in oligopoly markets. This will be performed by comparing the perfectly competitive market and the oligopoly market that all GenCos submit their bids in the form of linear supply function (LSF). To study the interactions between spot and forward markets, we extended the spot electricity market modeled by SFE model to include forward market. The two-settlement electricity market is used to analyze the GenCos' capacity withholding. Moreover, the effect of capacity withholding on distortion of the market clearing price is studied. It is widely accepted that the GenCos' capacity withholding depends significantly on many parameters such as the GenCos cost functions, the number of GenCos, the style of competition, the demand curve, and the demand elasticity. Another objective of this article is to examine a new index, distortion–withheld index (DWI), which could measure the potential ability of market for capacity withholding from these parameters point of view.

This article is organized as follows. The next section presents the strategic bidding model based on the SFE model in the spot electricity markets, taking into account the capacity withholding index, the capacity distortion index, and the price distortion index. In Section 3, we use a two-stage game model for assessing the mitigation effects of forward contracts on capacity withholding. Our proposed index, DWI, is formulated in Section 4. Section 5 is devoted to a three GenCo test system for assessing capacity withholding with formulated indices. In this section, another larger test system is also used and comparison between two test systems is presented. Our conclusions are stated in Section 6.

2. SPOT MARKET

2.1. Market model

Suppose there are N GenCos in the electricity market and each GenCo has a generator. Then the marginal cost of GenCo i will be as follows

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$$\begin{aligned} &\mathrm{MC}_i = a_i y_i + b_i \\ &a_i > 0 \end{aligned} \tag{1}$$

The aggregate demand function can be written as

$$\begin{aligned} &\pi = -\alpha Y + \xi \\ &\alpha > 0 \end{aligned} \tag{2}$$

where $Y = \sum_{i=1}^{N} y_i$ and there is negligible transmission loss.

In a perfectly competitive market, each GenCo should increase its generation up to the point where its marginal cost is equal to the market price as shown in the following equation:

$$\pi = a_i y_i + b_i \tag{3}$$

Because of the specific features of the electricity markets, the actual electricity markets may be better described in terms of oligopoly. Cournot and SFE models are used widely for oligopoly electricity market simulations. In this article, we used SFE model for oligopoly electricity market simulation. We assumed that GenCos submit their bids in the form of LSF. The supply function of each GenCo takes the following form

$$\begin{aligned} &y_i = \frac{1}{\hat{\alpha}_i}\left(\pi - \hat{\beta}_i\right) \\ &\hat{\alpha}_i > 0 \end{aligned} \tag{4}$$

Badlick [14] summarized the LSF parameterization into four classifications:

1) a-parameterization, where GenCo i can choose $\hat{\alpha}_i$ in Equation (4) arbitrarily but is required to specify a fixed and prechosen value of $\hat{\beta}_i$. The value of $\hat{\beta}_i$ is assumed to be equal to the value of b_i.
2) b-parameterization, where GenCo i can choose $\hat{\beta}_i$ in Equation (4) arbitrarily but is required to specify a fixed and prechosen value of $\hat{\alpha}_i$. The value of $\hat{\alpha}_i$ is assumed to be equal to the value of a_i.
3) $(a \propto b)$-parameterization, where GenCo i can choose $\hat{\alpha}_i$ and $\hat{\beta}_i$ subject to the condition that $\hat{\alpha}_i$ and $\hat{\beta}_i$ have a fix linear relationship. That is, there is a fixed prechosen positive k_i and GenCo i must then choose $\hat{\alpha}_i$ and $\hat{\beta}_i$ so that $\hat{\alpha}_i = k_i.a_i$ and $\hat{\beta}_i = k_i.b_i$.
4) (a, b)-parameterization, where GenCo i can choose $\hat{\alpha}_i$ and $\hat{\beta}_i$ arbitrarily.

According to this classification decision, variables are $\hat{\alpha}_i, \hat{\beta}_i$, and k_i. GenCo i, like other GenCos, seeks to maximize its profit (Ω_i) by changing its decision variable.

The profit function of each GenCo can be written as

$$\Omega_i = y_i \pi - C_i \tag{5}$$

where C_i is the cost function of GenCo i. The profit function can be differentiate with respect to price

$$\frac{\partial \Omega_i}{\partial \pi} = y_i + \pi \frac{\partial y_i}{\partial \pi} - \frac{\partial C_i}{\partial y_i}\frac{\partial y_i}{\partial \pi} = y_i + (\pi - \mathrm{MC}_i)\frac{\partial y_i}{\partial \pi} \tag{6}$$

According to Equations (2) and (4), y_i can be obtained by

$$y_i = \frac{\xi - \pi}{\alpha} - \sum_{j=1, j\neq i}^{N} \frac{\pi - \hat{\beta}_j}{\hat{\alpha}_j} \tag{7}$$

Substituting Equation (7) into Equation (6) and setting the derivative of GenCo i in Equation (6) as zero yields

$$y_i = (\pi - \mathrm{MC}_i)\hat{A}_i \tag{8}$$

where $\hat{A}_i = 1/\alpha + \sum_{j=1, i\neq j}^{N} 1/\hat{\alpha}_j$.

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2.2. Capacity withholding analyzing

This article uses the capacity withheld index ($\Delta y_i^{\text{withheld}}$), capacity distortion index ($\Delta y_i^{\text{distort}}$), and price distortion index ($\Delta\pi^{\text{distort}}$) to analyze the capacity withholding of GenCos in the oligopoly markets. These indices are defined as follows:

$$\Delta y_i^{\text{withheld}} = y_i^p(\pi^e) - y_i^e \tag{9}$$

$$\Delta y_i^{\text{distort}} = y_i^p - y_i^e \tag{10}$$

$$\Delta\pi^{\text{distort}} = \pi^e - \pi^p \tag{11}$$

Because of the capacity withholding, the output of GenCo i decreases to y_i^e and the market clearing price increases to π^e. $y_i^p(\pi^e)$ is the competitive output which would be produced at π^e by GenCo i.

To solve three problems that were mentioned earlier, we extended these indices and proposed a new integrated index, called DWI, in Section 4.

According to Equation (3), y_i^p can be obtained by

$$y_i^p(\pi^e) = \frac{\pi^e - b_i}{a_i} \tag{12}$$

By using Equation (8), the quantity generated by GenCo i under SFE model can be written as

$$\begin{aligned} y_i^e &= (\pi^e - \text{MC}_i)\hat{A}_i \\ &= \left(\pi^e - a_i y_i^e - b_i\right)\hat{A}_i \\ &= \frac{(\pi^e - b_i)\hat{A}_i}{1 + a_i\hat{A}_i} \end{aligned} \tag{13}$$

Substituting Equation (12) into Equation (13) yields

$$y_i^e = \frac{y_i^p(\pi^e)a_i\hat{A}_i}{1 + a_i\hat{A}_i} \tag{14}$$

Substituting Equation (14) into Equation (9) yields

$$\Delta y_i^{\text{withheld}} = \frac{y_i^e}{a_i\hat{A}_i} \tag{15}$$

The fraction of $\Delta y_i^{\text{withheld}}$ and y_i^e can be written as

$$\frac{\Delta y_i^{withheld}}{y_i^e} = \frac{1}{a_i\hat{A}_i} \tag{16}$$

According to Equation (15) in an oligopoly market with SFE model, $1/a_i\hat{A}_i$ of the quantity generated by each GenCo is withheld from the spot market. It can be observed that $\Delta y_i^{\text{withheld}}$ is affected by the slope of GenCos' marginal cost functions. It means that a GenCo with small slope of marginal cost function has more incentive to withhold the capacity. It can also be seen that $\Delta y_i^{\text{withheld}}$ depends on the value of $\hat{A}_i$. When α is increasing, the demand becomes less elastic and $\hat{A}_i$ becomes smaller and $\Delta y_i^{\text{withheld}}$ becomes larger. Moreover, $\Delta y_i^{\text{withheld}}$ is affected by the value of $\hat{\alpha}_i$ and the type of the LSF parameterization. According to $\hat{A}_i$ definition, if $\hat{\alpha}_i$ is larger than $\hat{\alpha}_{i'}$ (for any i and $i' \in 1, \ldots, N$), then $\hat{A}_i$ is larger than $\hat{A}_{i'}$ and then $\left(\Delta y_i^{\text{withheld}}/y_i^e\right) < \left(\Delta y_{i'}^{\text{withheld}}/y_{i'}^e\right)$ can be obtained.

According to Equation (3), y_i^p can be written as

$$y_i^p = \frac{\pi^p - b_i}{a_i} \tag{17}$$

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Substituting Equations (17) and (13) into Equation (10) yields

$$\Delta y_i^{distort} = \frac{\pi^p - b_i}{a_i} - \frac{(\pi^e - b_i)\hat{A}_i}{1 + a_i\hat{A}_i} = \frac{(\pi^p - b_i) - a_i\hat{A}_i(\pi^e - \pi^p)}{a_i\left(1 + a_i\hat{A}_i\right)} \tag{18}$$

Substituting Equations (11) and (17) into Equation (18) yields

$$\Delta y_i^{distort} = \frac{y_i^p - \hat{A}_i\Delta\pi^{distort}}{\left(1 + a_i\hat{A}_i\right)} = \frac{y_i^e + \Delta y_i^{distort} - \hat{A}_i\Delta\pi^{distort}}{\left(1 + a_i\hat{A}_i\right)} \tag{19}$$

Notice that $y_i^p = y_i^e + \Delta y_i^{\text{distort}}$. According to Equation (19), $\Delta y_i^{\text{distort}}$ can be obtained by

$$\Delta y_i^{\text{distort}} = \frac{1}{a_i}\left(\frac{y_i^e}{\hat{A}_i} - \Delta\pi^{\text{distort}}\right) \tag{20}$$

Equation (21) is obtained by using Equation (2)

$$\Delta\pi^{\text{distort}} = \pi^e - \pi^p = (-\alpha Y^e + \xi) - (-\alpha Y^p + \xi) = \alpha\Delta Y^{\text{distort}} \tag{21}$$

where $Y^p = \sum_{i=1}^{N} y_i^p$, $Y^e = \sum_{i=1}^{N} y_i^e$, and $\Delta Y^{\text{distort}} = Y^p - Y^e$ or $\Delta Y^{\text{distort}} = \sum_{i=1}^{N} \Delta y_i^{\text{distort}}$. Ideally, with the increase of $\Delta Y^{\text{distort}}$, the expected result should be the increment in $\Delta\pi^{\text{distort}}$. Equation (21) represents that $\Delta\pi^{\text{distort}}$ is influenced by the slope of the aggregate demand function. It means that in a market with a large slope of demand function, GenCos will have more incentive to lift up the market price.

Substituting Equation (21) into Equation (20) yields

$$\Delta y_i^{\text{distort}} = \frac{1}{a_i}\left(\frac{y_i^e}{\hat{A}_i} - \alpha\Delta Y^{\text{distort}}\right) \tag{22}$$

It can be observed that the value of $\Delta y_i^{\text{distort}}$ may be positive or negative. It depends on the slope of aggregate demand function, the slope of the GenCos' marginal cost functions and the LSF parameterization. In Section 4, we show that α/a_i ratio is important for measuring the ability of GenCos for capacity withholding. For GenCos with large α/a_i, $\Delta y_i^{\text{distort}}$ can be positive. This implies that the quantity generated by these GenCos has been decreased with respect to the perfect competition market. In fact, these GenCos can indirectly control the market clearing price and make the GenCos with small α/a_i become the marginal GenCos by capacity withholding. It may cause the GenCos with small α/a_i generate more than their perfect competitive generations (y_i^p), and for these GenCos, $y_i^e > y_i^p$ ($\Delta y_i^{\text{distort}} < 0$) may be obtained.

The total capacity distortion ($\Delta Y^{\text{distort}}$) can be also defined as the sum of the capacity distortions of all GenCos,

$$\Delta Y^{distort} = \sum_{i=1}^{N} \frac{1}{a_i}\left(\frac{y_i^e}{\hat{A}_i} - \alpha\Delta Y^{distort}\right) = \frac{\sum_{i=1}^{N}\left(\frac{y_i^e}{a_i\hat{A}_i}\right)}{1 + \sum_{i=1}^{N}\frac{\alpha}{a_i}} \tag{23}$$

The results described earlier can be used for the a-, b-, and ($a \propto b$)-parameterizations of LSF in the spot market. When GenCos use LSF with (a, b)-parameterization, the slope and the intercept of LSF of each GenCos are not unique, but the market clearing price and GenCos' outputs are unique and equal to the results of the Cournot model [27]. These results are summarized in Appendix A, and we use $\Delta y_i^{\text{withheld}}$, $\Delta y_i^{\text{distort}}$, and $\Delta\pi^{\text{distort}}$ of the Cournot model to analyze the capacity withholding of GenCos in SFE model with (a, b)-parameterization of LSF. According to Equation (A8) (see Appendix A) in this case, α/a_i of the generation power of each GenCo is withheld from the spot market.

Notice that $\Delta y_i^{\text{withheld}}$, $\Delta y_i^{\text{distort}}$, and $\Delta\pi^{\text{distort}}$ are comparison indices. In these indices, market outcomes in actual markets are compared with perfect competition. Therefore, we need to the perfect competition simulation. However, all of the formulations of the capacity withholding indices mentioned in this article

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just depend on y_i^e, a_i, α, and $\hat{\alpha}_i$. It means that there is no need to the perfect competition simulation when we use these indices for capacity withholding analyzing.

3. TWO-SETTLEMENT MARKET

In this section, a two-stage game model is used to formulate GenCos' strategic forward contracting and spot market competition in a two-settlement electricity market. In the first stage, GenCos sell their forward contracts in the forward market. In the forward market, it is assumed that all GenCos are economically rational and risk neutral, and there are enough risk neutral arbitrageurs in the market and the forward market price is an unbiased estimator of the spot market price. GenCo i solves the following problem to choose its forward market generation to maximize its profit

$$\max\left\{\Omega_i = \pi\left(y_i - y_i^c\right) + \pi_i^c y_i^c - C_i\right\}$$
$$s.t.$$
$$\pi_i^c = \pi \tag{24}$$

In this stage, the decision variable is y_i^c. In the second stage, GenCos compete with each other in the spot market by submitting their bids in the LSF form. In the second stage, the decision variable is y_i. According to Equation (8), the equation of this stage can be written as

$$y_i - y_i^c = (\pi - \mathrm{MC}_i)\hat{A}_i \tag{25}$$

The spot and forward decisions are taken sequentially, and the forward market equilibrium can be determined taking into account the effects of forward decisions on the spot market equilibrium. In the study if Zhang *et al*. [28], it is shown that the GenCos' marginal benefits of forward contracts are not larger than zero when they use the a-,b-, and ($a \propto b$)-parameterizations of LSF in the spot market and they have no incentive to enter the forward market. In this case, $y_i^c = 0$ and $\Delta y_i^{\text{withheld}}$, $\Delta y_i^{\text{distort}}$, and $\Delta\pi^{\text{distort}}$ are equal to Equations (15), (20), and (21), respectively.

It is also found that when GenCos use the (a, b)-parameterization of LSF in the spot market, $\hat{\alpha}_i$ and $\hat{\beta}_i$ for each GenCo are not unique but the GenCos' outputs and market clearing price are unique and equal to the results of a case that the GenCos use Cournot model in the spot market with strategic forward contracting in the forward market [28]. Then, we can use $\Delta y_i^{\text{withheld}}$, $\Delta y_i^{\text{distort}}$, and $\Delta\pi^{\text{distort}}$ of the Cournot model to analyze the capacity withholding of GenCos in the (a, b)-parameterization of LSF. These indices are formulated for the Cournot model in Appendix A.

When α is increasing, the demand becomes less elastic and the GenCos will have more incentive to withhold their capacities. According to Equation (A14) in Appendix A, with the increase of α, A_i becomes larger and the coefficient $1/(1+A_i)$ becomes smaller. Hence, the term $1/(1+A_i)$ can moderate the increase in the capacity withholding of GenCos due to the higher demand curve associated to the decreased demand elasticity.

4. DISTORTION–WITHHELD INDEX

In this section, the DWI has been proposed to assess the potential ability of market for capacity withholding. The DWI of a market is defined as $\Delta Y^{\text{distort}}$–$\Delta Y^{\text{withheld}}$ ratio. According to Equation (23), the DWI for an oligopoly market with SFE model can be written as

$$\mathrm{DWI} = \frac{\Delta Y^{distort}}{\Delta Y^{withheld}} = \frac{1}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{26}$$

where $\Delta Y^{\text{withheld}} = \sum_{i=1}^{N} \frac{y_i^e}{a_i A_i}$. DWI ranges from 0 to 1, and when this index is lower, it is more likely that the market has more potential ability for capacity withholding. In contrast, the higher DWI, the market is closer to competition.

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The DWI reflects the potential ability of the oligopoly market for capacity withholding as it only depends on the slope of aggregate demand function and the slope of GenCos' marginal cost functions. The realization of this potential ability depends on GenCos' strategies. In the spot market, all GenCos optimize their expected profit by their spot strategies (Cournot or SFE model) with or without forward market arrangement. Because of the different strategies or different market models, both $\Delta Y^{\text{withheld}}$ and $\Delta Y^{\text{distort}}$ are changed. However, according to Equations (26) and (A19), the $\Delta Y^{\text{distort}}$–$\Delta Y^{\text{withheld}}$ ratio (DWI) is constant (suppose the aggregate demand parameters in Equation (2) and the marginal cost parameters of each GenCo in Equation (1) are not changed).

Figure 1 shows the capacity withholding, capacity distortion and price distortion. To simplify the illustration, it is assumed in Figure ure 1 that there is only one GenCo in the market. Therefore, the subscript i is omitted from y_i^p, y_i^e, and $y_i^p(\pi^e)$. To assess the effect of marginal cost's slope on capacity withholding and DWI, we suppose that the slope of the GenCo's marginal cost function is decreased, as shown in Figure ure 1. Graphically, a marginal cost curve with lower slope (dash-dot line in Figure ure 1) will be located in lower position with respect to the original one (solid line in Figure 1). According to Equation (26), the DWI is decreased, and according to Equation (15), $\Delta Y^{\text{withheld}}$ is increased. In Figure 1, the increased capacity withheld is $\Delta Y^{'\text{ withheld}}$. This is represented in Figure 1 by the shift from strategic offer 1 to the strategic offer 2. It can be observed that because of the modification in supply curve, the price distortion is increased from $\Delta\pi^{\text{distort}}$ to $\Delta\pi^{'\text{ distort}}$. The capacity distortion is increased form $\Delta Y^{\text{distort}}$ to $\Delta Y^{'\text{ distort}}$. Therefore, the decrease in DWI is coherent with decreasing in competition and increasing in capacity withholding.

According to Equation (26), the share of each GenCo in price distortion ($\Delta\pi_i^{\text{distort}}$) is

$$\Delta\pi^{\text{distort}} = \alpha\ \text{DWI}\ \Delta Y^{\text{withheld}} = \alpha\ \text{DWI}\ \sum_{i=1}^{N}\left(\frac{\alpha}{\alpha_i}y_i^e\right) = \sum_{i=1}^{N}\Delta\pi_i^{\text{distort}} \tag{27a}$$

$$\Delta\pi_i^{\text{distort}} = \alpha\ \text{DWI}\Delta y_i^{\text{withheld}} = \alpha\ \text{DWI}\frac{\alpha}{a_i}y_i^e \tag{27b}$$

Intertemporal constraints such as production ramp rate limits can affect the market power and market equilibrium. When the GenCos are forced to follow the intertemporal constraints, the market price can increase and the generation of the GenCos will change over the time horizon, particularly during peak demand hours. We assumed a three-period market as shown in Figure 2. Assume that ramp rate limits are not binding in periods 1 and 2. $\Delta y_{i,1}^{\text{distort}}$ and $\Delta y_{i,2}^{\text{distort}}$ are the capacity distortion

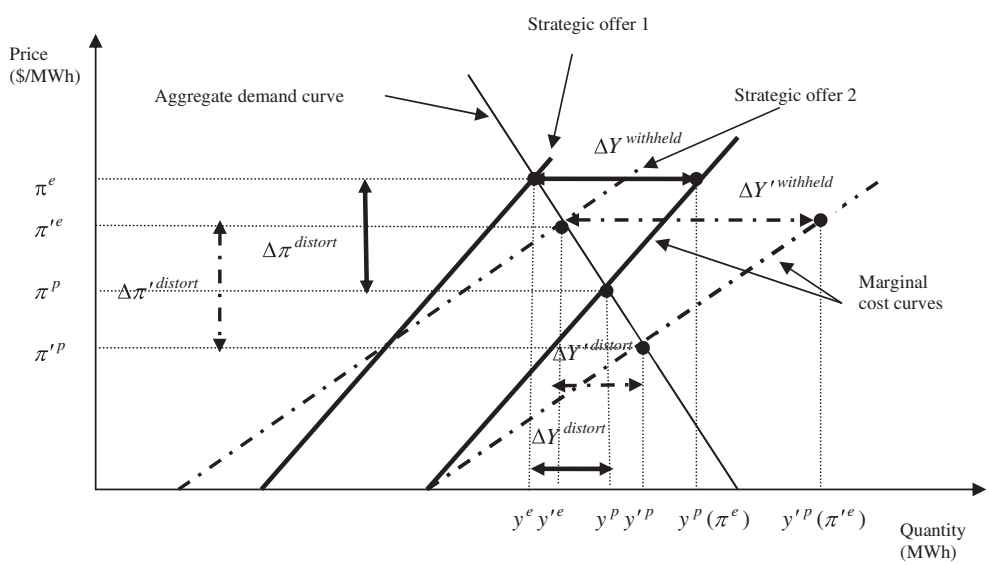

Figure 1. Illustration of the effect of the decreased marginal cost's slope on the capacity withholding of market and DWI.

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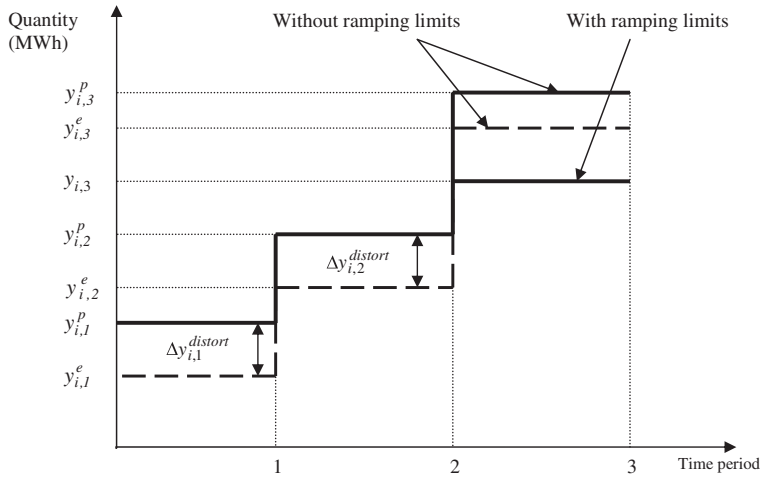

Figure 2. The effect of intertemporal constraints on capacity withholding.

of GenCo i in periods 1 and 2, respectively. Assume that ramp up rate limit is binding in period 3. This intertemporal constraint provides the GenCo i's generation under both perfect competition and oligopoly. GenCo i is forced to reduce producing energy in period 3 when the ramp up limit is imposed and there is no capacity withholding. However, because of the limited generation, the market price turns into far higher value.

Market power may be exercised by GenCos' strategic maintenance in electricity markets. In an oligopolistic power market, GenCos' strategic maintenance is a way of capacity withholding. Each GenCo maximizes its profit by strategically putting its capacity on maintenance as well as withholding generation from available capacity. GenCos' strategic maintenance has an intertemporal aspect, that is, performing maintenance during certain weeks/months has effects on availability of other weeks/months and hence maintenance/generation of those months [21].

5. NUMERICAL STUDIES

We present the analytical study of capacity withholding of GenCos with reference to a three GenCo test system in [28]. There are two cases for this test system. In case E, there is the SFE model in the spot market competition without the forward contracts. The four different LSF parameterization cases mentioned in Section 2 are considered in cases E_1 (a-parameterization), E_2 (b-parameterization), E_3 (($a \propto b$)-parameterization), and E_4 ((a, b)-parameterization). The SFE-type competition in the spot market with the strategic forward contracts is considered in case F. In case F_1, there is (a, b)-parameterization in the spot market with forward market arrangement. The quadratic cost data of system are given in Table I. The aggregate demand parameters in Equation (2) are $\alpha = 2$ \$/(MWh·GWh) and $\xi = 90$\$/MWh. In the study of Zhang *et al*. [28], a coevolutionary genetic algorithm is used to determine market equilibrium and optimal strategies of GenCos (coefficients of LSF) in the two-settlement market with different simulation cases. This algorithm contains an inner loop and an outer loop. The optimal strategies in the spot market and forward market are determined in the inner loop and outer loop, respectively. These loops are described in a procedure in [28].

Table I. Cost parameters.

GenCo no.	1	2	3
Cost parameter a_i\$/(MWh·GWh)	1	1.5	2
Cost parameter b_i\$/MWh	10	12	8

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The simulation results of cases E_1, E_2, E_3, E_4, and F_1 are shown in Table II, and the values of $\hat{A}_i$ and DWI are listed in Table III.

The capacity withheld index, the capacity distortion index, and the price distortion index of cases E_1, E_2, E_3, E_4, and F_1 are listed in Table IV.

According to Table III, the DWI in all cases is 0.187. It means that the type of the LSF parameterization does not influence the $\Delta Y^{\text{distort}}$–$\Delta Y^{\text{withheld}}$ ratio although the values of $\Delta Y^{\text{distort}}$ and $\Delta Y^{\text{withheld}}$ are affected. The capacity withholding indices in Table IV ($\Delta y_i^{\text{withheld}}$, $\Delta y_i^{\text{distort}}$, and $\Delta \pi^{\text{distort}}$) show that in all cases, a GenCo with small slope of marginal cost function (GenCo 1) have more incentive to withhold the capacity in the market. According to Table IV when GenCos use (a, b)-parameterization in the spot market, there is the least competitive case. From the results for case E_4, it is shown that the capacity withholding of the GenCos will be lead to a higher market price distortion in this case with respect to other cases. Because of the forward contract, the values of the capacity withholding indices are decreased in case F_1. It can be seen that when GenCos use b-parameterization in the spot market, the market is more competitive. This is because the $\Delta y_i^{\text{withheld}}$ of the GenCos depends significantly on the slope of their marginal cost function. It can be observed that the capacity distortion of the GenCo 3 in cases E_1, E_2, and E_3 is negative. It means that this GenCo prefers to generate more than its competitive output in these cases.

Table II. Simulation results for cases E_1, E_2, E_3, E_4, and F_1.

Case	GenCo no.	y_i (GWh)	y_i^c (GWh)	$\hat{\alpha}_i$ (\$/(MWh·GWh)	$\hat{\beta}_i$ (\$/MWh)	π (\$/MWh)
E_1	1	10.97	–	1.75	12	31.21
	2	9.68	–	2.19	10	
	3	8.74	–	2.65	8	
E_2	1	11.19	–	1	18.71	29.91
	2	9.95	–	1.5	14.97	
	3	8.90	–	2	12.10	
E_3	1	11.04	–	1.32	21.04	30.61
	2	9.82	–	1.85	12.37	
	3	8.84	–	2.38	9.53	
E_4	1	9.11	–	–	–	39.34
	2	8.38	–	–	–	
	3	7.83	–	–	–	
F_1	1	10.59	5.48	–	–	32.82
	2	9.42	5.07	–	–	
	3	8.58	4.75	–	–	

Table III. $\hat{A}_i$, A_i, and DWI of cases E_1, E_2, E_3, E_4, and F1.

Case	GenCo no.	A_i	$\hat{A}_i$ (\$/(MWh·GWh))$^{-1}$	DWI
E_1	1	–	1.333	0.187
	2	–	1.448	
	3	–	1.527	
E_2	1	–	1.666	0.187
	2	–	2.000	
	3	–	2.166	
E_3	1	–	1.458	0.187
	2	–	1.672	
	3	–	1.791	
E_4	1	–	–	0.187
	2	–	–	
	3	–	–	
F_1	1	1.071	–	0.187
	2	1.166	–	
	3	1.238	–	

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Table IV. Capacity withholding indices of cases E_1, E_2, E_3, E_4, and F_1.

Case	GenCo no.	$\Delta y_i^{\text{withheld}}$ (GWh)	$\Delta Y^{\text{withheld}}$ (GWh)	$\Delta y_i^{\text{distort}}$ (GWh)	$\Delta Y^{\text{distort}}$ (GWh)	$\Delta\pi^{\text{distort}}$ (\$/MWh)
E_1	1	8.229	15.546	2.406	2.911	5.82
	2	4.456		0.574		
	3	2.861		−0.049		
E_2	1	6.716	12.086	2.190	2.263	4.52
	2	3.316		0.298		
	3	2.054		−0.208		
E_3	1	7.571	13.954	2.345	2.613	5.22
	2	3.915		0.433		
	3	2.467		−0.145		
E_4	1	18.220	37.223	4.262	6.979	13.96
	2	11.173		1.868		
	3	7.830		0.851		
F_1	1	10.226	19.871	2.776	3.725	7.45
	2	5.812		0.832		
	3	3.833		0.108		

To investigate the effects of the demand function's slope and cost parameters on the capacity withholding, cases F_2, F_3, F_4, and F_5 are performed. These cases are the same as case F_1 expect for in case F_2, $\alpha = 4$, in case F_3, $\alpha = 1$, in case F_4, $a_1 = 2$, and in case F_5, $b_1 = 8$. The simulation results are listed in Table V. The values of A_i and DWI are summarized in Table VI, and the capacity withholding indices are shown in Table VII.

When comparing cases F_2 and F_3 with case F_1, it can be observed that the DWI is decreased and the capacity withholding indices are increased with the increased slope of the demand function (see Tables VI and VII). The promising fact is that a small value of the incremental elasticity added to case F_3 will remarkably bring down the $\Delta Y^{\text{distort}}$, $\Delta Y^{\text{withheld}}$, and $\Delta\pi^{\text{distort}}$, resulting in the increased DWI from 0.187 to 0.315.

In case F_4, the ability of GenCo 1 for capacity withholding is the same as GenCo 3. However, $y_1 < y_3$ and $\Delta y_1^{\text{withheld}} < \Delta y_3^{\text{withheld}}$. In this case, the DWI is increased with respect to case F_1, and $\Delta Y^{\text{distort}}$, $\Delta Y^{\text{withheld}}$, and $\Delta\pi^{\text{distort}}$ are decreased and the GenCos become more competitive in case F_4 with respect to case F_1. The DWI of case F_5 is the same as case F_1. It means that the DWI is not affected by the intercept of GenCos' marginal cost functions. According to Equation (26), the DWI is only affected by the slope of GenCos' marginal cost functions and demand function's slope. Notice that GenCo 1 in case F_5 becomes cheaper than case F_1 and the output of market in case F_5 is slightly higher than case F_1. Therefore, $\Delta Y^{\text{distort}}$, $\Delta Y^{\text{withheld}}$, and $\Delta\pi^{\text{distort}}$ in case F_1 are slightly smaller than case F_5.

Table V. Simulation results for cases F_2, F_3, F_4, and F_5.

Case	GenCo no.	y_i (GWh)	y_i^c (GWh)	$\hat{\alpha}_i$ (\$/(MWh·GWh)	$\hat{\beta}_i$ (\$/MWh)	π (\$/MWh)
F_2	1	12.31	7.17	–	–	44.88
	2	11.17	6.64	–	–	
	3	10.29	6.22	–	–	
F_3	1	7.66	3.24	–	–	24.08
	2	6.89	3.13	–	–	
	3	6.37	3.02	–	–	
F_4	1	7.89	4.08	–	–	35.41
	2	10.16	5.08	–	–	
	3	9.24	4.78	–	–	
F_5	1	12.04	6.23	–	–	31.67
	2	8.94	4.82	–	–	
	3	8.18	4.53	–	–	

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Table VI. $\hat{A}_i$, A_i, and DWI for cases F2, F3, F4, and F5.

Case	GenCo no.	A_i	$\hat{A}_i$ (\$/(MWh·GWh))$^{-1}$	DWI
F_2	1	1.393	–	0.103
	2	1.466	–	
	3	1.527	–	
F_3	1	0.733	–	0.315
	2	0.833	–	
	3	0.900	–	
F_4	1	1.071	–	0.230
	2	1.000	–	
	3	1.071	–	
F_5	1	1.071	–	0.187
	2	1.166	–	
	3	1.238	–	

Table VII. Capacity withholding indices for cases F_2, F_3, F_4, and F_5.

Case	GenCo no.	$\Delta y_i^{\text{withheld}}$ (GWh)	$\Delta Y^{\text{withheld}}$ (GWh)	$\Delta y_i^{\text{distort}}$ (GWh)	$\Delta Y^{\text{distort}}$ (GWh)	$\Delta\pi^{\text{distort}}$ (\$/MWh)
F_2	1	20.576	40.798	3.696	4.220	16.88
	2	12.078		0.825		
	3	8.144		−0.295		
F_3	1	4.420	8.598	1.705	2.715	2.715
	2	2.502		0.695		
	3	1.676		0.318		
F_4	1	3.809	15.043	0.338	3.471	5.22
	2	6.773		2.145		
	3	4.461		0.990		
F_5	1	11.627	20.785	3.833	3.897	7.79
	2	5.503		0.307		
	3	3.655		−0.241		

The effects of capacity constraints on the capacity withholding are also studied, and the simulation results, the DWI, and the capacity withholding indices are shown in Tables VIII, IX, and X, respectively. Cases E_5, E_6, and E_7 are the same as cases E_1, E_2, and E_3, except for considering the capacity constraint of GenCo 1 and setting its output to 5 GWh.

It can be seen that when GenCo 1 hits its capacity constraints, the DWI increases. When comparing cases E_1, E_2, and E_3 with E_5, E_6, and E_7, respectively, it is clear that $\Delta\pi^{\text{distort}}$ and $\Delta Y^{\text{distort}}$ are higher and $\Delta Y^{\text{withheld}}$ is lower when GenCo 1 hits its capacity constraints. Capacity withholding indices are shown in Figures 3–6.

Table VIII. Simulation results for cases E_5, E_6, and E_7.

Case	GenCo no.	y_i (GWh)	y_i^c (GWh)	$\hat{\alpha}_i$ (\$/(MWh·GWh)	$\hat{\beta}_i$ (\$/MWh)	π (\$/MWh)
E_5	1	5	–	–	–	39.02
	2	10.65	–	2.72	10	
	3	9.84	–	3.15	8	
E_6	1	5	–	–	–	37.44
	2	10.98	–	1.5	20.97	
	3	10.30	–	2	16.83	
E_7	1	5	–	–	–	38.52
	2	10.73	–	2.21	14.76	
	3	10.01	–	2.75	11.00	

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Table IX. $\hat{A}_i$, A_i, and DWI for cases E_5, E_6, and E_7.

Case	GenCo no.	A_i	$\hat{A}_i$ (\$/(MWh·GWh))$^{-1}$	DWI
E_5	1	–	–	0.300
	2	–	0.817	
	3	–	0.867	
E_6	1	–	–	0.300
	2	–	1	
	3	–	1.166	
E_7	1	–	–	0.300
	2	–	0.863	
	3	–	0.951	

Table X. Capacity withholding indices for cases E_5, E_6, and E_7.

Case	GenCo no.	$\Delta y_i^{\text{withheld}}$ (GWh)	$\Delta Y^{\text{withheld}}$ (GWh)	$\Delta y_i^{\text{distort}}$ (GWh)	$\Delta Y^{\text{distort}}$ (GWh)	$\Delta \pi^{\text{distort}}$ (\$/MWh)
E_5	1	–	14.636	–	4.300	8.60
	2	8.689		2.950		
	3	5.674		1.373		
E_6	1	–	11.759	–	3.520	7.04
	2	7.320		2.625		
	3	4.439		0.918		
E_7	1	–	13.550	–	4.056	8.11
	2	8.288		2.880		
	3	5.262		1.206		

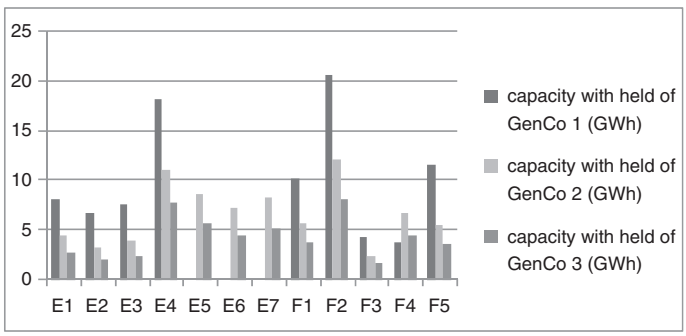

Figure 3. Capacity withheld index for three GenCos study case.

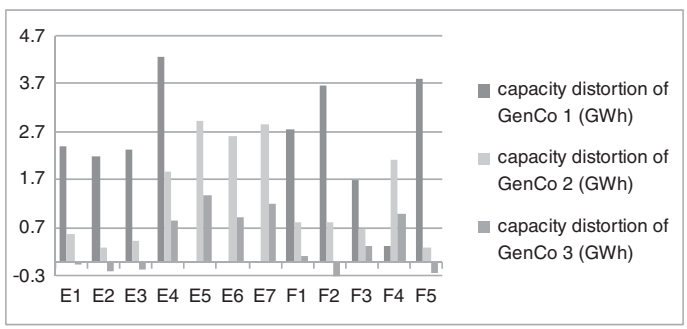

Figure 4. Capacity distortion index for three GenCos study case.

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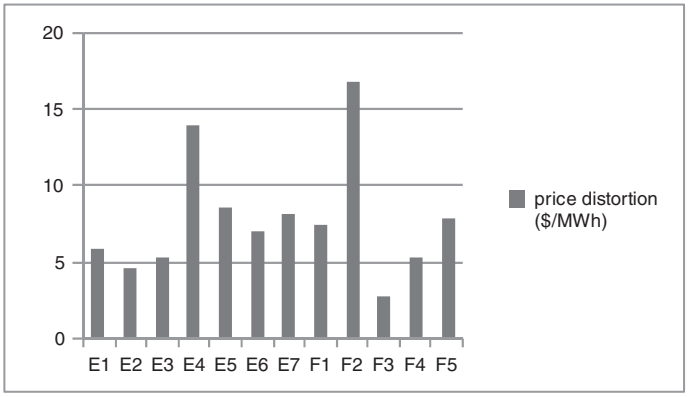

Figure 5. Price distortion index for three GenCos study case.

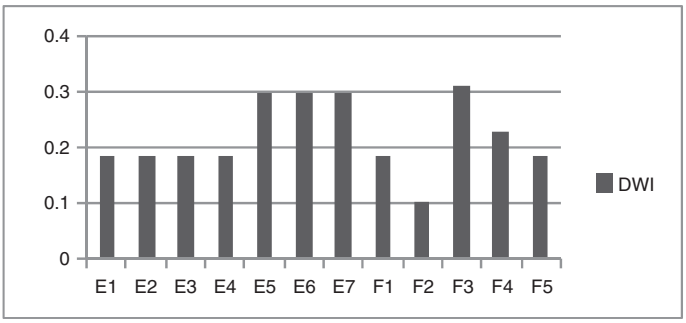

Figure 6. DWI for three GenCos study case.

We apply the obtained indices to a five GenCos test system to obtain some insights on the outcomes of larger system and to show the applicability to real systems. In this market, α =10£/(MWh·GWh) and ξ =350£/MWh. The cost parameters of five GenCos, which are based on the cost data for the five strategic firms in England and Wales after 1999 [27], are shown in Table XI. The simulation results and capacity withholding indices for (a, b)-parameterization in five GenCos test system (case G) are listed in Table XII.

According to Tables III, IV, and XII, the values of DWI for cases E_4 and G are 0.18 and 0.05, respectively. It means that market power and capacity withholding in the three GenCos system are lower than the five GenCos system. Moreover, the values of $\Delta\pi^{\text{distort}}$ for cases E_4 and G are 13.96$/

Table XI. Cost parameters.

GenCo no.	1	2	3	4	5
Cost parameter a_i£/(MWh·GWh)	2.687	4.615	1.789	1.930	4.615
Cost parameter b_i£/MWh	12	12	8	8	12

Table XII. Capacity withholding indices for cases G.

Case	GenCo no.	y_i (GWh)	$\Delta y_i^{\text{withheld}}$ (GWh)	$\Delta Y^{\text{withheld}}$ (GWh)	$\Delta y_i^{\text{distort}}$ (GWh)	$\Delta Y^{\text{distort}}$ (GWh)	$\Delta\pi^{distort}$ (£/MWh)	DWI
G	1	5.39	20.05	106.07	0.14	5.35	53.5	0.05
	2	4.68	10.13		−1.45			
	3	6.14	34.31		4.41			
	4	6.07	31.45		3.73			
	5	4.68	10.13		−1.45			

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MWh and 53.5£/MWh. It means that the effect of capacity withholding on market price in case E_4 is lower than case G.

6. CONCLUSION

A linear SFE model with four different LSF parameterizations is presented to analyze GenCos' capacity withholding in oligopolistic electricity markets. Depending on the LSF parameterization, GenCos' capacity withholding may be decreased or increased. Market power mitigation effects of strategic forward contracting are evaluated with a set of comparison indices that allow to measure and analyze capacity withholding comparing the oligopoly outcomes with ideal perfect competition. We conclude that price distortion is dependent not only on the slope of system demand function but also on total capacity distortion in the market. The DWI can provide a quantitative way for market designers or regulators to monitor the overall potential ability of market for capacity withholding. The results for SFE and Cournot models show that this index is not affected by the type of the market modeling. It is also found that GenCos' capacity withholding depends significantly on the slope of their marginal cost function and the slope of system demand function. We believe that the term α/a_i can be used as a criterion to determine which player is more strategic than other players, and it can be an alternative for the Federal Energy Regulatory Council recommendation. It can also be extended to analyze capacity withholding when collusion with GenCos is considered in future works.

7. LIST OF SYMBOLS

π	Spot market price
π_i^c	Forward contract price for GenCo i
π^e	Oligopoly market price
π^p	Perfect competition market price
α	Slope of aggregate demand function
ξ	Intercept of aggregate demand function
C_i	Cost function for GenCo i
MC_i	Marginal cost function for GenCo i
Ω_i	Profit function for GenCo i
a_i	Slope of marginal cost function for GenCo i
b_i	Intercept of marginal cost function for GenCo i
y_i	Quantity generated by GenCo i
y_i^c	Quantity generated by GenCo i in forward market
Y	Total generation
$\hat{\alpha}_i, \hat{\beta}_i$	Coefficients of LSF for GenCo i
y_i^P	Competitive generation for GenCo i
y_i^e	Oligopoly generation for GenCo i
$y_i^p(\pi^e)$	Competitive generation at π^e for GenCo i
Y^p	Total competitive generation
Y^e	Total oligopoly generation
t	Time period, $t = 1, \ldots T$
$y_{i,t}$	Quantity generated by GenCo i in t
$y_{i,t}^p$	Competitive generation for GenCo i in t
$y_{i,t}^e$	Oligopoly generation for GenCo i in t
S_i	Market share for GenCo i
ε	Demand elasticity

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APPENDIX A

When there is the Cournot model, the optimization problem can be written as

$$\Omega_i = \left(y_i - y_i^c\right)\pi - C_i + \pi_i^c y_i^c \tag{A1a}$$

$$\frac{\partial \Omega_i}{\partial y_i} = \left(y_i - y_i^c\right)\frac{\partial \pi}{\partial y_i} + \pi - \mathrm{MC}_i \tag{A1b}$$

By setting the derivative of GenCo i's profit with respect to its decision variable (y_i) as zero, Equation (A1b) can be written as

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DOI: 10.1002/etep

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$$\pi + \left(y_i - y_i^c\right)\frac{\partial \pi}{\partial y_i} = \mathrm{MC}_i \tag{A2}$$

Equation (A2) can be written as

$$\pi\left(1 - \frac{S_i}{|\varepsilon|}\right) = \mathrm{MC}_i \tag{A3}$$

According to Equation (A3), y_i^e can be written as

$$y_i^e = \frac{\pi^e\left(1 - \frac{S_i}{|\varepsilon|}\right) - b_i}{a_i} \tag{A4}$$

where

$$S_i = \frac{y_i^e - y_i^c}{\frac{\xi}{\alpha} - \frac{\pi^e}{\alpha}} \tag{A5}$$

$$\varepsilon = \frac{\partial Y}{\partial \pi} \times \frac{\pi}{Y} = -\frac{1}{\alpha} \times \frac{\pi^e}{\frac{\xi}{\alpha} - \frac{\pi^e}{\alpha}} \tag{A6}$$

Substituting Equations (A4) and (12) into Equation (9) yields

$$\Delta y_i^{\text{withheld}} = \frac{\pi^e \frac{S_i}{|\varepsilon|}}{a_i} \tag{A7}$$

Substituting Equations (A5) and (A6) into Equation (A7) yields

$$\Delta y_i^{\text{withheld}} = \frac{\alpha}{a_i}\left(y_i^e - y_i^c\right) \tag{A8}$$

Substituting Equations (A4) and (17) into Equation (10) yields

$$\Delta y_i^{\text{distort}} = \left(y_i^e - y_i^c - \Delta Y^{\text{distort}}\right)\frac{\alpha}{a_i} \tag{A9}$$

The total capacity distortion index can be written as

$$\Delta Y^{\text{distort}} = \frac{\sum_{i=1}^{N} \frac{\alpha}{a_i}\left(y_i^e - y_i^c\right)}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A10}$$

$$\Delta \pi^{\text{distort}} = \alpha \Delta Y^{\text{distort}} = \alpha \times \frac{\sum_{i=1}^{N} \frac{\alpha}{a_i}\left(y_i^e - y_i^c\right)}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A11}$$

The DWI can be obtained by

$$\mathrm{DWI} = \frac{\Delta Y^{distort}}{\Delta Y^{withheld}} = \frac{1}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A12}$$

If there is no forward contract, then y_i^c=0 (for any $i \in 1, \ldots N$). For strategic forward contracting, the relationship between y_i^c and y_i can be determined by [16]

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$$y_i^c = \frac{A_i}{1 + A_i} y_i \tag{A13}$$

where

$$A_i = \sum_{\substack{j=1 \\ j \neq i}}^{N} \left(\frac{\alpha}{\alpha + a_j} \right) \tag{A14}$$

By substituting Equation (A13) into Equations (A8), (A9), (A10), and (A11), the following equations can be obtained for strategic forward contracting

$$\Delta y_i^{\text{withheld}} = \frac{\alpha}{a_i} y_i^e \frac{1}{1 + A_i} \tag{A15}$$

$$\Delta y_i^{\text{distort}} = \frac{\alpha}{a_i} \left(\frac{1}{1 + A_i} y_i^e - \Delta Y^{\text{distort}} \right) \tag{A16}$$

$$\Delta Y^{\text{distort}} = \frac{\sum_{i=1}^{N} \frac{\alpha}{a_i} \left(\frac{1}{1+A_i} y_i^e \right)}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A17}$$

$$\Delta \pi^{\text{distort}} = \alpha \Delta Y^{\text{distort}} = \alpha \times \frac{\sum_{i=1}^{N} \frac{\alpha}{a_i} \left(\frac{1}{1+A_i} y_i^e \right)}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A18}$$

$$\text{DWI} = \frac{\Delta Y^{distort}}{\Delta Y^{withheld}} = \frac{1}{1 + \sum_{i=1}^{N} \frac{\alpha}{a_i}} \tag{A19}$$

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